

RAKYAT ELECTRONIC SYSTEM

Data Structure & Algorithm PROJECT

JUNE
2021

GROUP MEMBERS

NAME	ROLES
MOHD HAMKA	Team Leader & System Development
FAIZATUL FITRI	Documentation & System Support
HARITH ZAKWAN	Application Development
IMAN TARMIZI	Video Presentation

INTRODUCTION

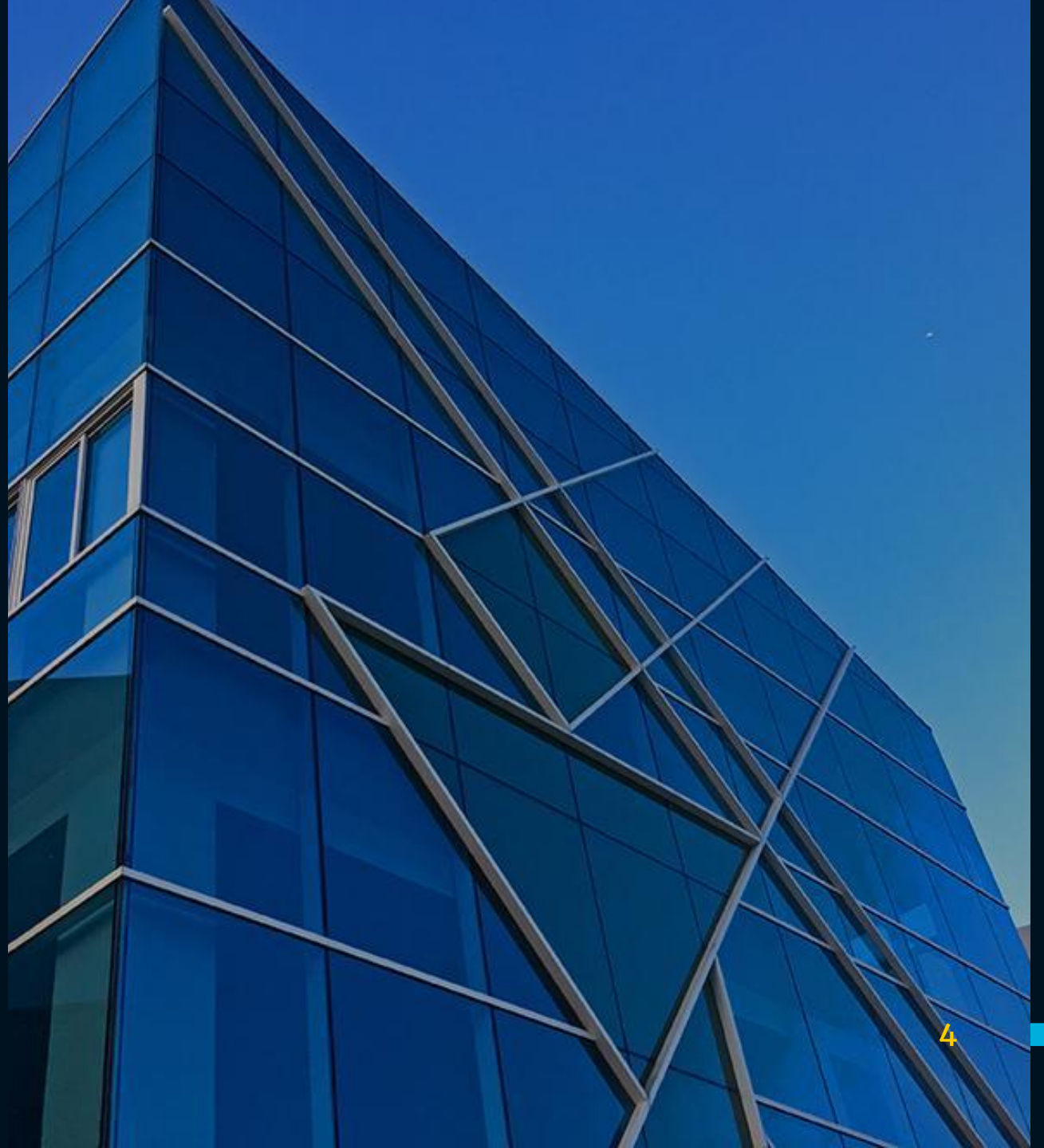
INTRODUCTION

Rakyat Electronic System

This project aims to develop a **Sales & Inventory Management System** for Rakyat Electronic System to efficiently store, manage, and retrieve weekly and monthly sales records. The system will utilize data structures and algorithms to organize product information, including product model, price, quantity, and sales data.

The system will manage various electronic products such as mobile phones, computer accessories, and other related items. By applying appropriate data management techniques, the system will allow Rakyat Electronic System to track sales performance, update stock availability, and maintain organized product records efficiently.

Through this system, the company can improve inventory control, reduce manual record-keeping efforts, and make informed decisions based on accurate sales information.



RELATED TOPICS

- Stack
- Queue
- Search Algorithm
- Sort Algorithm

WHY DO WE NEED TO APPLY THIS TOPIC

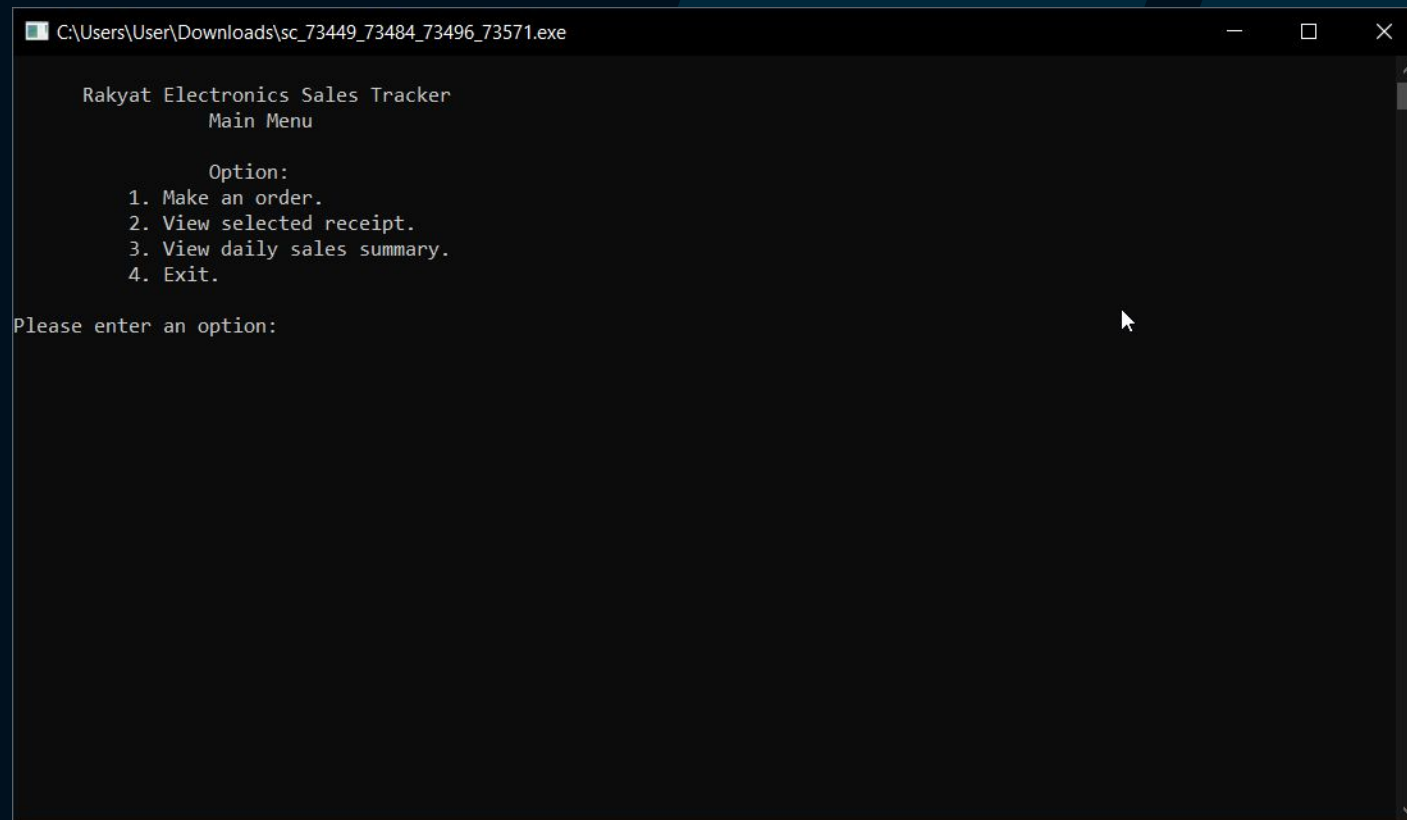
STACK OR QUEUE

- A **stack** is a linear data structure that follows the **Last In First Out (LIFO)** principle, where the latest added element is removed first. In this project, a stack can be used to store recent sales transactions, allowing the system to easily access the latest records.
- A **queue** follows the **First In First Out (FIFO)** principle, where the first added element is processed first. In the Rakyat Electronic System, a queue can be used to manage customer orders by processing requests based on the order they are received.

SEARCH OR SORT ALGORITHM

- A **search algorithm** is used to find specific data stored in a system. In this project, searching algorithms help users find product information such as product name, model, or category quickly.
- A **sorting algorithm** arranges data in a specific order. In the Rakyat Electronic System, sorting can be used to organize products based on price, quantity, or sales records, making data management easier and more efficient.

OUR SYSTEM VIEW



```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

Rakyat Electronics Sales Tracker
Main Menu

Option:
1. Make an order.
2. View selected receipt.
3. View daily sales summary.
4. Exit.

Please enter an option:
```

OUR SYSTEM VIEW

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

Rakyat Electronics Sales Tracker
Main Menu

Option:
1. Make an order.
2. View selected receipt.
3. View daily sales summary.
4. Exit.

Please enter an option:
1
```

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

Category:
a. Mobile Phones
b. Computer & Desktops
c. Home Entertainment
d. Kitchen Appliances
e. Health & Beauty

Any items in categories above can be added.

Option:
1. Add item details.
2. Edit item details.
3. Delete selected item.
4. View receipt.
5. Return to main menu.

Please enter an option:
```

OUR SYSTEM VIEW

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

Category:
a. Mobile Phones
b. Computer & Desktops
c. Home Entertainment
d. Kitchen Appliances
e. Health & Beauty

Any items in categories above can be added.

Option:
1. Add item details.
2. Edit item details.
3. Delete selected item.
4. View receipt.
5. Return to main menu.

Please enter an option: 1
```

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

-----
Rakyat Electronics
Receipt
-----
Receipt No: 6
-----
No.  Model Type                Price (RM)  Quantity  Subtotal (RM)
-----
1
-----
Total (RM):      0.00
-----
Receipt currently have no existing items.
Enter model type:
```

OUR SYSTEM VIEW

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

Category:
a. Mobile Phones
b. Computer & Desktops
c. Home Entertainment
d. Kitchen Appliances
e. Health & Beauty

Any items in categories above can be added.

Option:
1. Add item details.
2. Edit item details.
3. Delete selected item.
4. View receipt.
5. Return to main menu.

Please enter an option: 2_
```

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

-----
Rakyat Electronics
Receipt
-----
Receipt No: 6
-----
No.  Model Type                Price (RM)  Quantity  Subtotal (RM)
-----
1    iPHONG                    100.00     12        1200.00
-----
Total (RM): 1200.00
-----

Receipt have 4 remaining item(s).

Enter no. of item to be edited: 1

Edit option:
1. Model Type.
2. Price.
3. Quantity.
4. All of the above.

Please enter an option:
```

OUR SYSTEM VIEW

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

Category:
a. Mobile Phones
b. Computer & Desktops
c. Home Entertainment
d. Kitchen Appliances
e. Health & Beauty

Any items in categories above can be added.

Option:
1. Add item details.
2. Edit item details.
3. Delete selected item.
4. View receipt.
5. Return to main menu.

Please enter an option: 3
```

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

-----
Rakyat Electronics
Receipt
-----
Receipt No: 6
-----
No.  Model Type                Price (RM)  Quantity  Subtotal (RM)
-----
1    oPPA                      123.00     21         2583.00
-----
Total (RM):  2583.00
-----

Receipt have 4 remaining item(s).

Enter no. of item to be deleted: 1
```

OUR SYSTEM VIEW

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

Category:
a. Mobile Phones
b. Computer & Desktops
c. Home Entertainment
d. Kitchen Appliances
e. Health & Beauty

Any items in categories above can be added.

Option:
1. Add item details.
2. Edit item details.
3. Delete selected item.
4. View receipt.
5. Finalize order and return to main menu.

Please enter an option: 4
```

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

-----
Rakyat Electronics
Receipt
-----
Receipt No: 6
-----
No.  Model Type                Price (RM)  Quantity  Subtotal (RM)
-----
1    iPHONG                    100.00     12        1200.00
-----
2    oPPA                      123.00     21        2583.00
-----
Total (RM):  3783.00
-----

Receipt have 3 remaining item(s).
Press any key to continue . . .
```

OUR SYSTEM VIEW

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

Rakyat Electronics Sales Tracker
Main Menu

Option:
1. Make an order.
2. View selected receipt.
3. View daily sales summary.
4. Exit.

Please enter an option: 2
```

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

-----
Rakyat Electronics
Receipt
-----
Receipt No: 6
-----
No.  Model Type                Price (RM)  Quantity  Subtotal (RM)
-----
1    iPHONG                    100.00     12         1200.00
-----
2    oPPA                      123.00     21         2583.00
-----
Total (RM):    3783.00
-----

Continue viewing receipts? (Y/N):
```

OUR SYSTEM VIEW

```
C:\Users\User\Downloads\sc_73449_73484_73496_73571.exe

Rakyat Electronics Sales Tracker
Main Menu

Option:
1. Make an order.
2. View selected receipt.
3. View daily sales summary.
4. Exit.

Please enter an option: 3
```

Rakyat Electronics
Daily Sales Summary

No.	Model Type	Price (RM)	Quantity	Subtotal (RM)
1	Acer Nitro 5	2999.00	2	5998.00
2	Apple MacBook Pro	5599.00	2	11198.00
3	Apple TV	899.00	2	1798.00
4	Asus ROG Phone 5	3199.00	7	22393.00
5	Asus TUF Gaming	3499.00	2	6998.00
6	Google Pixel 4a	2199.00	2	4398.00
7	Huawei Vision 5	2999.00	4	11996.00
8	Kenwood Sandwich Maker	149.00	9	1341.00
9	Khind Hairdryer H01400	39.00	5	195.00
10	LG UR74 Series	4249.00	3	12747.00
11	Lenovo Legion 5i 15	4449.00	4	17796.00
12	MSI GF63 Thin	3499.00	1	3499.00
13	Nidea Jug Kettle	119.00	2	238.00
14	Oral-B Power Toothbrush	39.00	4	156.00
15	Panasonic Blender	159.00	3	477.00
16	Panasonic Shaver	69.00	7	483.00
17	Panasonic Electric Oven	299.00	8	2392.00
18	Panasonic Hair Straightener	279.00	3	837.00
19	Phillip Rice Cooker	379.00	5	1895.00
20	Phillips Essential Curler	69.00	3	207.00
21	Samsung A9000 4K UHD	4999.00	2	9998.00
22	Samsung Galaxy S20	3399.00	4	13596.00
23	Sony PlayStation 5	9999.00	1	9999.00
24	Xiaomi Mi 11	2999.00	2	5998.00
25	IPADMG	100.00	12	1200.00
26	iPhone 12 Pro	4899.00	3	14697.00
27	OPPO	123.00	21	2583.00
Total (RM):				165113.00

Option:
1. Sort by Model Type.
2. Sort by Subtotal Price.
3. Return to main menu.

Please enter an option:

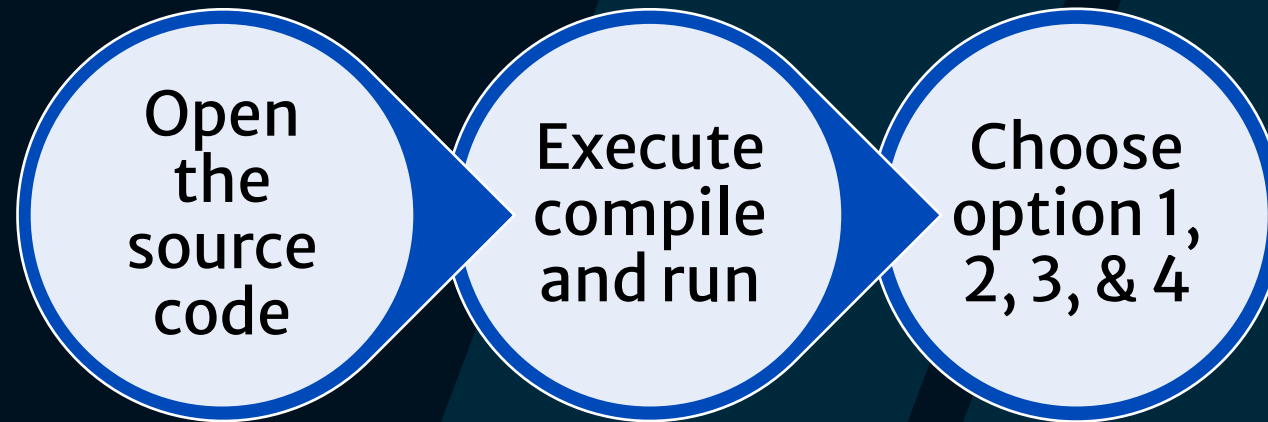
A close-up, slightly angled shot of a silver laptop resting on a dark-stained wooden desk. The laptop is open, showing a black keyboard and a large, rectangular trackpad. The wood grain of the desk is prominent, running diagonally from the bottom left towards the top right. The lighting is soft and even, highlighting the textures of the wood and the metallic finish of the laptop. The word "DESIGN" is superimposed in a white, serif font, centered horizontally and partially overlapping the laptop's trackpad and the desk surface.

DESIGN

A low-angle, upward-looking photograph of a modern skyscraper with a glass facade. The building's structure, including its windows and metal framework, creates a strong sense of height and perspective, converging towards the top of the frame. The sky is a vibrant blue with scattered white clouds. The overall image has a clean, architectural feel.

USER MANUAL

USER MANUAL



User Manual

- **Option 1: CREATE ORDER**

In this option, users can create and manage an order. After creating an order, users will be provided with additional options such as **Add**, **Edit**, **Delete**, and **View**. Users can perform these actions with a maximum limit of **5 items** before returning to the main menu.

- **Option 2: CHECK RECEIPT**

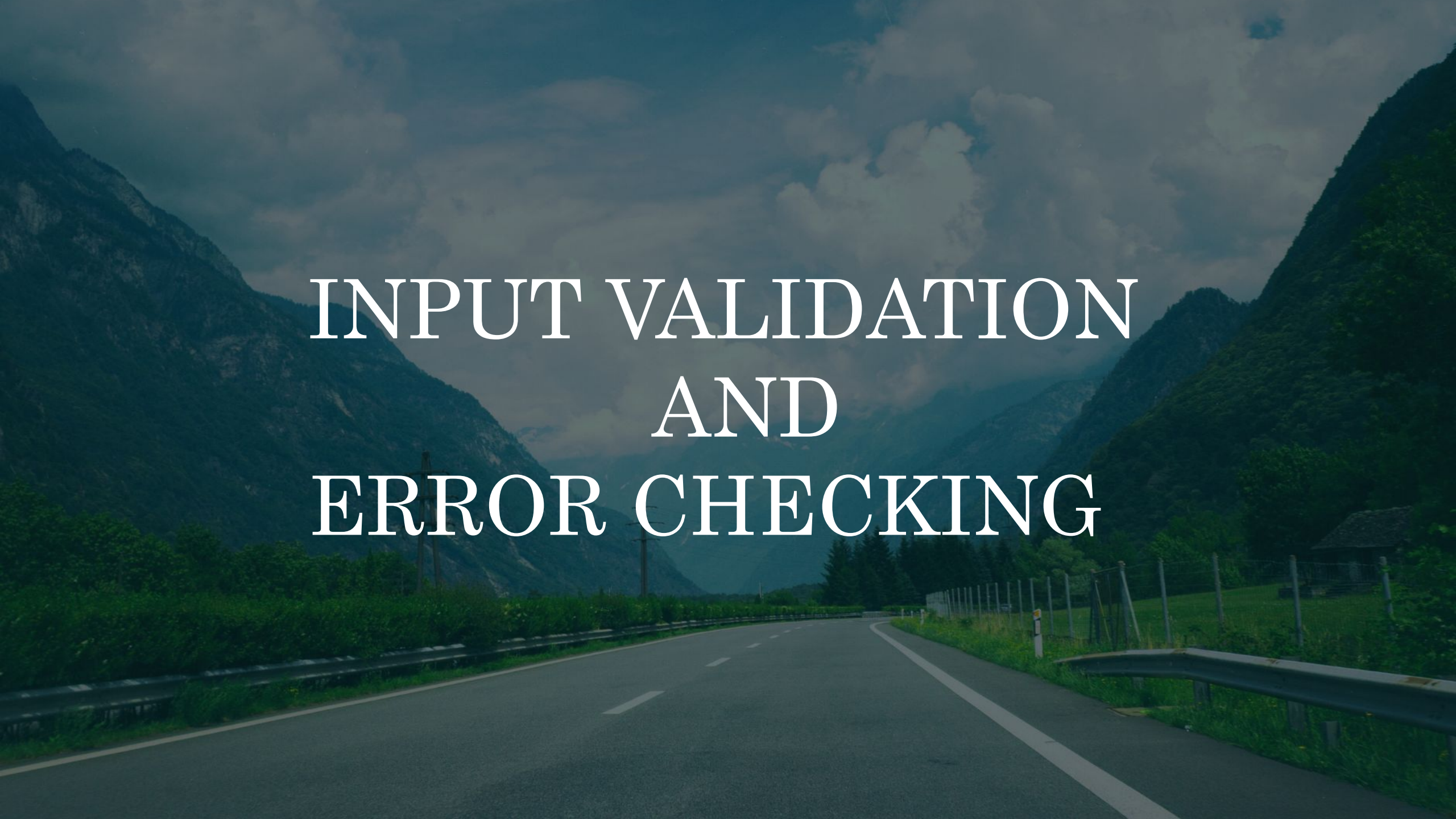
In this option, users can view a receipt by entering the corresponding receipt number. Users must provide a valid receipt number to proceed. If users want to view the preset receipt data, they can enter **receipt number 5**.

- **Option 3: VIEW DAILY SALES SUMMARY**

In this option, users can view the daily sales summary of the system. Users will be provided with additional options to sort the sales records based on **model type** or **subtotal price**. After viewing or sorting the sales data, users can return to the main menu.

- **Option 4: EXIT SYSTEM**

In this option, the system will redirect users to the exit page and automatically terminate the program. This allows users to safely close the application after completing their tasks.

A scenic mountain road with a road sign. The road is paved and curves through a valley. On the left, there is a road sign with a white background and a black border, featuring a black silhouette of a person walking. The road is flanked by green grass and shrubs. In the background, there are steep, forested mountains under a cloudy sky. The overall scene is a peaceful mountain landscape.

INPUT VALIDATION AND ERROR CHECKING



THANK YOU!
